ARJAN SINGH GILLAN

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SUMMARY

Biomedical engineering student with hands-on eye-care device experience, having run verification and validation on an AR vision-assistance platform for macular degeneration. Skilled in device design (CAD/FEA), material characterization, testing, and data analysis, with a strong interest in ophthalmic and vision-care R&D.

EDUCATION

University of California, Irvine — Irvine, CA
B.S. in Biomedical Engineering | GPA: 3.64 — *Graduation: June 2028*

EXPERIENCE

Assistant Project Manager, University Rover Challenge — Irvine, CA
Legacy Robotics at UC Irvine — *April 2026 – Present*

- Lead an 80-person, 5-subteam engineering organization and manage a ~$15K budget, coordinating cross-functional design reviews and integration under competition deadlines.
- Developed an onboard soil biosignature assay to detect protein concentrations and authored the chemical safety and lab-handling protocol for onboard testing.

Applications & Clinical Engineering Intern — Irvine, CA
Eyedaptic — AR/AI Vision Assistance for Macular Degeneration — *Jan 2026 – June 2026*

- Ran verification and validation testing of an AI-driven vision-assistance platform on AR glasses for patients with macular degeneration, benchmarking 4 AR headsets and 4 AI agents on response quality and camera performance.
- Authored 2 technical documents of comparative device evaluations that guided product and engineering improvements.

Undergraduate Researcher, Biomechanics — Irvine, CA
UCI Rehabilitation & Augmentation Lab — *April 2025 – Present*

- Conduct human-subjects gait rehabilitation trials, analyzing 3D motion-capture data in Vicon Nexus to extract joint kinematics and step metrics across baseline and stimulation sessions.
- Use MATLAB to quantify how movement direction and speed affect ankle proprioception, informing rehabilitation protocols for stroke-patient recovery.

Undergraduate Researcher, Neuroimaging — Irvine, CA
UCI Center for Neural Circuit Mapping — Xu Lab — *Jan 2025 – Present*

- First-author on an NIH-funded project building a novel neuroanatomical brain atlas (222 regions) by segmenting MRI scans in ITK-SNAP, and developed a program to quantify regional brain shrinkage for Alzheimer's research.
- Trained on the LifeCanvas protocol and completed histology training — sectioning, antibody staining, and microscopy.

TECHNICAL PROJECTS

Bridgeless Eyewear System — Irvine, CA
Ophthalmic Device Design & FEA Project — *May 2026 – June 2026*

- Co-designed a bridgeless eyewear system in SolidWorks that redistributes frame load off the nasal bridge for post-surgical and congenital nasal conditions.
- Validated a 58.9 g 3D-printed prototype through static FEA (stress, strain, displacement) and motion studies, iterating ball-detent sizing and a spring-loaded slide for adjustable fit.

Autocorrelogram Firing Program — Irvine, CA
Data Analysis Project — *Nov 2025 – Dec 2025*

- Built a custom MATLAB tool to classify neuronal firing patterns with autocorrelogram visualization of electrophysiological time-series datasets, automating a previously manual workflow to improve repeatability.

LEADERSHIP

Vice President of Finance — Irvine, CA
Engineering Student Council at UCI — *April 2026 – Present*

- Oversee a ~$40K annual budget for a 22-member board, managing financial planning across council programs and events.

Engineering Ambassador — Irvine, CA
The Samueli School of Engineering, UC Irvine — *March 2026 – Present*

- Represent the School of Engineering for the Undergraduate Student Affairs Office, supporting recruitment through outreach to prospective students and families.

SKILLS & INTERESTS

Technical: Testing & Validation (V&V), Material Characterization, Microscopy & Histology, Design Verification, CAD & FEA (SolidWorks), Materials & Prototyping (3D Printing, CNC), Data & Time-Series Analysis, MATLAB, C++, C#, Arduino, ITK-SNAP, Vicon Nexus, Technical Documentation

Interests: Event Planning, Public Speaking, Cooking, Rock Climbing, Golf, Tennis